

Markov decision processes

Lecture 3 — MDPs, slides 1–55. Book chapter 4.

Definition 1. Markov state

A state representation is *Markov* if for every t , every history h_t and every action a_t ,

$$p(s_{t+1} \mid s_t, a_t) = p(s_{t+1} \mid h_t, a_t).$$

The future is conditionally independent of the past given the present.

Definition 2. Markov decision process

A discounted infinite-horizon MDP is a tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$: a state set, an action set, transition dynamics $p(s' \mid s, a)$ satisfying Definition 1, a reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, R_{\max}]$, and a discount factor $\gamma \in [0, 1)$. Unless stated otherwise $\mathcal{S}$ and $\mathcal{A}$ are finite.

Lemma 3. Values are finite, and how finite

For every policy and every state, $0 \leq V^\pi(s) \leq \frac{R_{\max}}{1 - \gamma}$.

Proof.

Rewards lie in $[0, R_{\max}]$, so $0 \leq \sum_h \gamma^h r_h \leq R_{\max} \sum_{h \geq 0} \gamma^h = R_{\max}/(1 - \gamma)$, the geometric series converging because $\gamma < 1$. Expectations preserve the bounds.

Definition 4. Discounted state-action distribution

$$d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^\pi(s, a; s_0).$$

For an initial distribution μ , write $d_\mu^\pi(s, a) = \mathbb{E}_{s_0 \sim \mu}[d_{s_0}^\pi(s, a)]$.

Lemma 5. It is a distribution, and that is the point of the $(1 - \gamma)$

$d_{s_0}^\pi$ is a probability distribution over $\mathcal{S} \times \mathcal{A}$, and for every bounded f ,

$$\mathbb{E}^\pi \left[\sum_{h=0}^{\infty} \gamma^h f(s_h, a_h) \mid s_0 \right] = \frac{1}{1 - \gamma} \mathbb{E}_{(s,a) \sim d_{s_0}^\pi} [f(s, a)].$$

Proof.

Non-negativity is clear. Summing over (s, a) and exchanging the (absolutely convergent) sums,

$$\sum_{s,a} d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_h \gamma^h \sum_{s,a} \mathbb{P}_h^\pi(s, a; s_0) = (1 - \gamma) \sum_h \gamma^h = 1,$$

since $\mathbb{P}_h^\pi(\cdot; s_0)$ is a distribution for each h . The identity is the same exchange applied to f .

Theorem 6. Bellman expectation equations

For every policy π and all s, a :

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot | s)} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} [V^\pi(s')] \right], \\ Q^\pi(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} [V^\pi(s')], \\ V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot | s)} [Q^\pi(s, a)]. \end{aligned}$$

For a deterministic policy the first reads $V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, \pi(s))} [V^\pi(s')]$ and the last $V^\pi(s) = Q^\pi(s, \pi(s))$.

Proof.

Split the return after the first step, using $G_0 = r(s_0, a_0) + \gamma G_1$:

$$V^\pi(s) = \mathbb{E}^\pi[G_0 \mid s_0 = s] = \mathbb{E}^\pi[r(s_0, a_0) \mid s_0 = s] + \gamma \mathbb{E}^\pi[G_1 \mid s_0 = s].$$

Condition the second term on (a_0, s_1) and use the tower rule. By Definition 1 and the fact that π depends only on the current state, the conditional distribution of the trajectory from s_1 onwards given (s_0, a_0, s_1) is the same as the distribution of a trajectory started at s_1 ; hence $\mathbb{E}^\pi[G_1 \mid s_0 = s, a_0 = a, s_1 = s'] = V^\pi(s')$. Averaging over $s' \sim p(\cdot \mid s, a)$ and then over $a \sim \pi(\cdot \mid s)$ gives the first identity; the argument for Q^π is the same with a_0 fixed.

Definition 7. Optimal value and optimal policy

$V^*(s) = \sup_\pi V^\pi(s)$ and $Q^*(s, a) = \sup_\pi Q^\pi(s, a)$. A policy π^* is optimal if $V^{\pi^*}(s) = V^*(s)$ for every s .

Theorem 8. Bellman optimality equation

$$\begin{aligned} V^*(s) &= \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} [V^*(s')] \right], \\ Q^*(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} \left[\max_{a'} Q^*(s', a') \right]. \end{aligned}$$

Theorem 9. Theorem 1 — greedy with respect to Q^* is optimal

Let $\hat{\pi}(s) \in \arg \max_a Q^*(s, a)$. Then $V^{\hat{\pi}} = V^*$, and therefore $\pi^* = \hat{\pi}$ is an optimal policy.

Proof.

By Theorem 8, $V^* = \mathcal{B}V^*$ and $Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} [V^*(s')]$. Since $\hat{\pi}(s)$ maximises $Q^*(s, \cdot)$,

$$\begin{aligned} (\mathcal{B}V^*)(s) &= \max_a Q^*(s, a) = Q^*(s, \hat{\pi}(s)) \\ &= r(s, \hat{\pi}(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, \hat{\pi}(s))} [V^*(s')] = (\mathcal{B}^{\hat{\pi}}V^*)(s), \end{aligned}$$

so V^* is a fixed point of $\mathcal{B}^{\hat{\pi}}$. By Theorem 6 so is $V^{\hat{\pi}}$, and both are bounded by Lemma 3.

A bounded fixed point of $\mathcal{B}^{\hat{\pi}}$ is unique. If u and v are two, then for every s

$$|u(s) - v(s)| = \gamma \left| \mathbb{E}_{s'} [u(s') - v(s')] \right| \leq \gamma \|u - v\|_\infty,$$

because the rewards cancel, so $\|u - v\|_\infty \leq \gamma \|u - v\|_\infty$. The norm is finite and $\gamma < 1$, hence $u = v$.

Therefore $V^{\hat{\pi}} = V^*$, and $\hat{\pi}$ attains the supremum of Definition 7 at every state.

Theorem 10. Theorem 2 — the optimality equation has one solution

If $V : \mathcal{S} \rightarrow \mathbb{R}$ satisfies $V(s) = \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} V(s')]$ for every s , then $V = V^*$.

Proof.

Both V and V^* satisfy the same equation, so for every s ,

$$\begin{aligned}
 |V(s) - V^*(s)| &= \left| \max_a [r(s, a) + \gamma \mathbb{E}_{s'} V(s')] \right. \\
 &\quad \left. - \max_a [r(s, a) + \gamma \mathbb{E}_{s'} V^*(s')] \right| \\
 &\leq \max_a \left| [r(s, a) + \gamma \mathbb{E}_{s'} V(s')] \right. \\
 &\quad \left. - [r(s, a) + \gamma \mathbb{E}_{s'} V^*(s')] \right| \\
 &= \max_a \gamma \left| \mathbb{E}_{s' \sim p(\cdot|s,a)} [V(s') - V^*(s')] \right| \\
 &\leq \gamma \max_a \mathbb{E}_{s' \sim p(\cdot|s,a)} |V(s') - V^*(s')|.
 \end{aligned}$$

The first inequality is $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$; the last is Jensen. Applying the same bound to $|V(s') - V^*(s')|$ and iterating k times,

$$|V(s) - V^*(s)| \leq \gamma^k \max_{a_0, \dots, a_{k-1}} \mathbb{E}_{s_k} |V(s_k) - V^*(s_k)| \leq \gamma^k \|V - V^*\|_\infty,$$

and since $\gamma < 1$ and $\|V - V^*\|_\infty$ is finite, the right-hand side tends to 0. Hence $|V(s) - V^*(s)| = 0$ for every s .